Scratch Agent Test Report

Session: 20251004_171710

Generated: 2025-10-04 17:26:50

Overall Statistics

TOTAL TESTS
34

ROUTING ACCURACY
94.12%

RESPONSE ACCURACY
76.47%

AVG EXECUTION TIME
11.77s

Category Statistics

MAKE BLOCKS
85.71%

6/7 routing correct | Response: 85.71% (6/7) | Time: 13.49s

CODE EXPLAIN
87.5%

7/8 routing correct | Response: 87.5% (7/8) | Time: 11.0s

CODE DEBUGGING

100.0%

8/8 routing correct | Response: 25.0% (2/8) | Time: 11.91s

GIVE INSTRUCTIONS

100.0%

8/8 routing correct | Response: 100.0% (8/8) | Time: 13.07s

GENERAL AGENT

100.0%

3/3 routing correct | Response: 100.0% (3/3) | Time: 5.97s

Make Blocks Tests

Test 1 ✓

Prompt: Create blocks to move the sprite 10 steps when green flag clicked

Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 13.01s

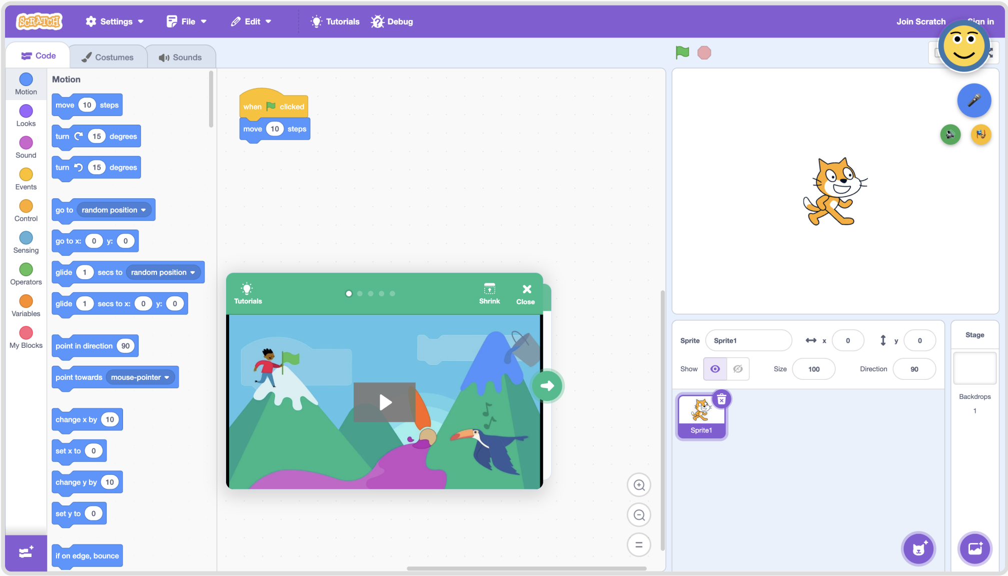
Agent Response:

Okay, I'm ready to turn those techy instructions into super fun and easy steps for kids! Just paste the message here, and I'll make it awesome. ✨

Response Verification: Accurate
Route Correct: True
Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:
Blocks Found: True | Workspace Length: 210 characters

View Full Workspace Data




```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. when Code block. Block Coordinates: (X: 350, Y: 143 )
 2. move Code block. Block Coordinates: (X: 350, Y: 175 )
```

Test 2 ✗

Prompt: Show me how to make the sprite say hello

Expected Route: make_blocks | **Actual Route:** give_instructions | **Time:** 8.31s

Agent Response:

Hey there, Scratch Explorers! Let's make your sprite talk! It's super easy and fun. 1. **Start with a Green Flag:** * Go to "Events" (it's in the blocks on the left side of the screen). * Grab the "when green flag clicked" block. It's like the "go" button for your code! 2. **Make Your Sprite Say Something:** * Now, go to "Looks" (also on the left, where the blocks are). * Find the "say message" block and snap it right under the "when green flag clicked" block. * See the word "Hello!" in the "say message" block? Click on it and type whatever you want your sprite to say! It should look like this: ```scratch when green flag clicked say [Hello!] ``` Now, click the green flag above the stage. Ta-da! Your sprite will say "Hello!" or whatever cool message you typed in! Isn't that awesome?

Response Verification: Not Accurate
Route Correct: False
Notes: Wrong route: expected make_blocks, got give_instructions

Test 3 ✓

Prompt: I want you to make blocks that turn right 15 degrees and move 10 steps

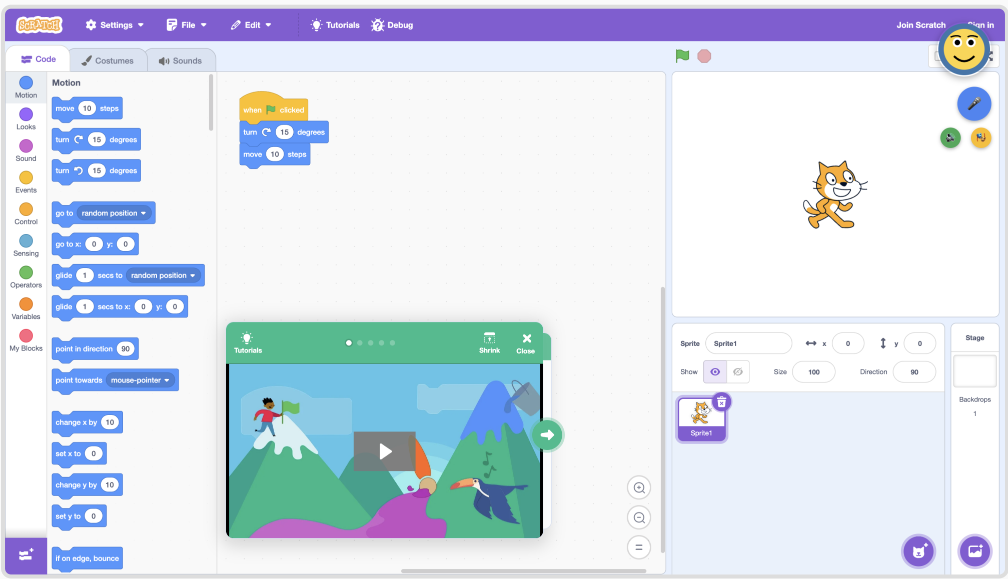
Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 13.01s

Agent Response:

Okay, I'm ready to turn those techy instructions into super fun steps for kids! Let's make Scratch programming easy and exciting! Just give me the message!

Response Verification: Accurate
Route Correct: True
Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:
Blocks Found: True | Workspace Length: 269 characters
View Full Workspace Data



```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. when Code block. Block Coordinates: (X: 350, Y: 143 )
 2. turn Code block. Block Coordinates: (X: 350, Y: 175 )
 3. move Code block. Block Coordinates: (X: 350, Y: 208 )
```

Test 4 ✓

Prompt: Help me create blocks when the green flag clicked repeatedly move 10 steps while making a sound

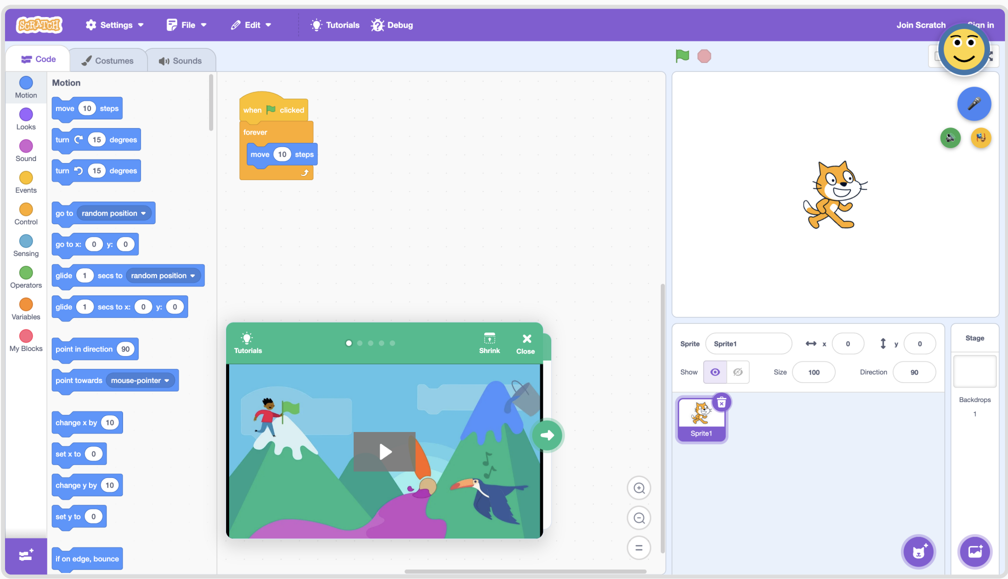
Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 13.62s

Agent Response:

Okay, I'm ready to turn those techy instructions into super fun and easy steps for kids! Just paste the message here, and I'll make it awesome!

Response Verification: Accurate
Route Correct: True
Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:
Blocks Found: True | Workspace Length: 272 characters
View Full Workspace Data



```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. when Code block. Block Coordinates: (X: 350, Y: 143 )
 2. forever Code block. Block Coordinates: (X: 350, Y: 175 )
 3. move Code block. Block Coordinates: (X: 361, Y: 208 )
```

Test 5 ✓

Prompt: Create blocks that say hello for 2 seconds, then hide and show the sprite

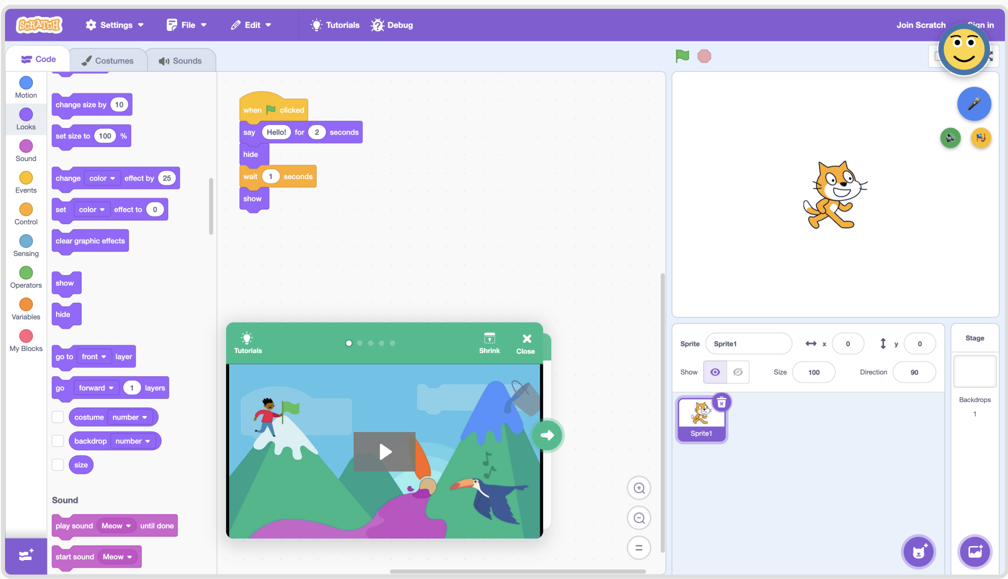
Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 18.17s

Agent Response:

Okay, I'm ready to turn techy instructions into super fun and easy steps for kids! Just give me the message, and I'll make it awesome!

Response Verification: Accurate
Route Correct: True
Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:
Blocks Found: True | Workspace Length: 386 characters
View Full Workspace Data



```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. when Code block. Block Coordinates: (X: 350, Y: 143 )
 2. say Code block. Block Coordinates: (X: 350, Y: 175 )
 3. hide Code block. Block Coordinates: (X: 350, Y: 208 )
 4. wait Code block. Block Coordinates: (X: 350, Y: 240 )
 5. show Code block. Block Coordinates: (X: 350, Y: 272 )
```

Test 6

✓

Prompt: Make a program that repeats moving 10 steps three times

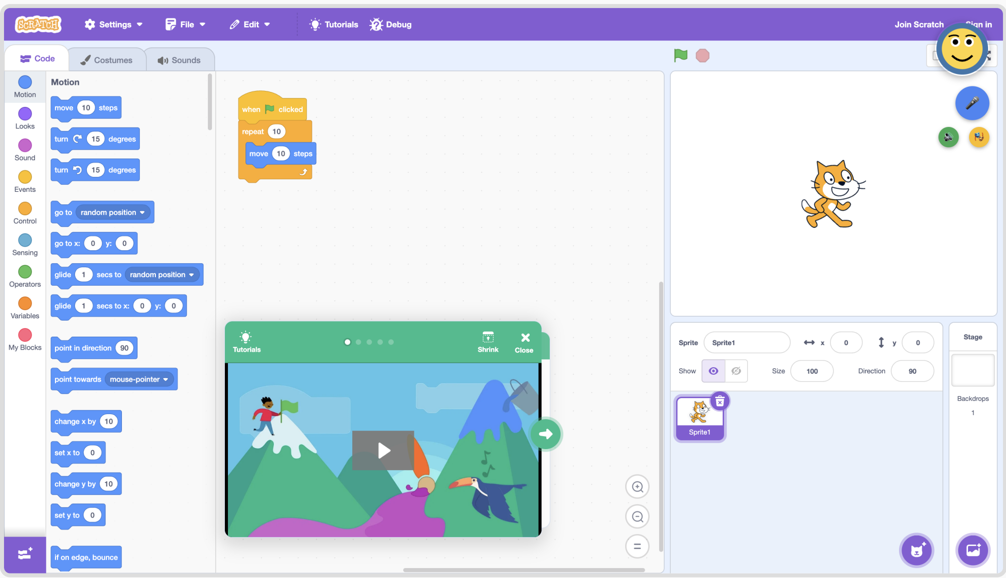
Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 12.78s

Agent Response:

Okay, I'm ready to turn those techy instructions into super fun steps for kids! Let's make Scratch programming easy and exciting! ✨

Response Verification: Accurate
Route Correct: True
Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:
Blocks Found: True | Workspace Length: 271 characters
View Full Workspace Data



```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. when Code block. Block Coordinates: (X: 350, Y: 143 )
 2. repeat Code block. Block Coordinates: (X: 350, Y: 175 )
 3. move Code block. Block Coordinates: (X: 361, Y: 208 )
```

Test 7 ✓

Prompt: Show me blocks to reset the timer, repeat some actions, and play a sound

Expected Route: make_blocks | **Actual Route:** make_blocks | **Time:** 15.55s

Agent Response:

Okay, I'm ready to turn those techy instructions into super fun and easy steps for kids! Just paste the message here, and I'll make it awesome!

Response Verification: Accurate

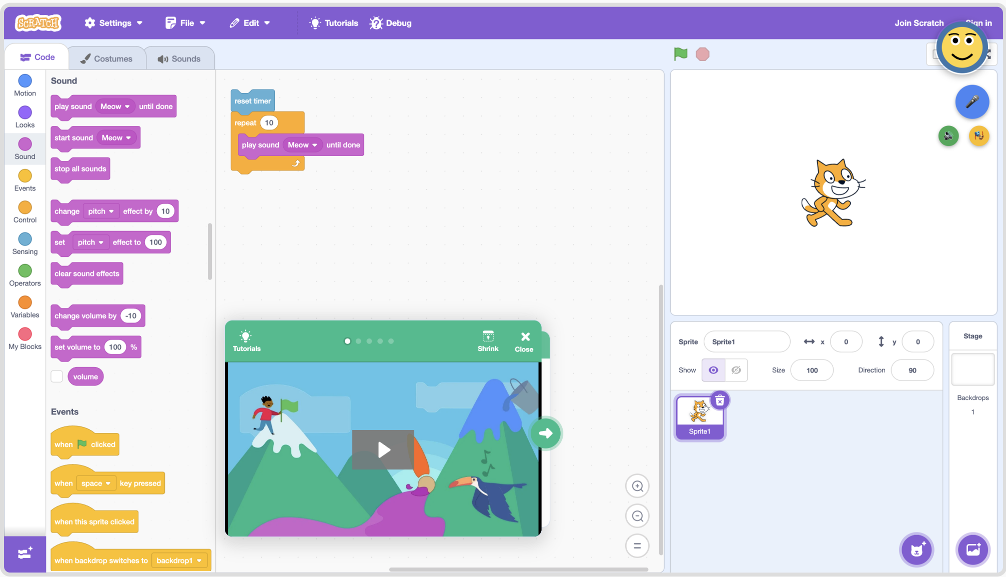
Route Correct: True

Notes: Contains block creation instructions, Make_blocks route - lenient accuracy

Blocks Test Results:

Blocks Found: True | Workspace Length: 153 characters

View Full Workspace Data



```
 Code space start from coordinates (X: 310, Y: 160). List of used
blocks in the code space:
 1. repeat Code block. Block Coordinates: (X: 339, Y: 164 )
```

Code Explain Tests

Test 1 ✓

Prompt: What does my current code do?

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 9.67s

Agent Response:

Hey there, Scratch superstar! I see you have a "repeat" block in your coding area. It's like a magic loop that can do things over and over! * **"Repeat" Block:** You can find this block in the "Control" section. It's now at X: 339 and Y: 164 on your screen. The "repeat" block helps you repeat actions. Whatever blocks you put *inside* it will happen again and again! Right now, your "repeat" block is empty, so it's not doing anything. Let's add some action! Drag other blocks *inside* the "repeat" block. Then, watch your code come to life as it repeats those actions! You've got this!

Response Verification: Accurate
Route Correct: True
Notes: Contains code explanation, Addresses prompt (1/4 key words match, 25.0%)

Test 2 ✓

Prompt: Explain the blocks I've placed in the workspace

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 10.43s

Agent Response:

Hey there, Scratch explorer! Let's see what cool blocks you've got! 1. **`when [green flag] clicked`**: This block is like the "go" button for your game! When you click the green flag, your whole project starts from here. It's at the top, like a hat! 2. **`repeat [10]`**: This block is like a magic loop! It tells Scratch to do something again and again. The number 10 means it will repeat whatever you put inside it ten times! It's like a container that holds other blocks. **So, here's the big picture:** You've got a "go" button (the green flag) and a "repeat" loop. Right now, the loop is empty, so it's not doing anything yet. But don't worry! You can put other blocks *inside* the "repeat" block to make awesome things happen over and over! What fun things will you make it do?

Response Verification: Accurate
Route Correct: True
Notes: Contains code explanation, Addresses prompt (1/5 key words match, 20.0%)

Test 3 ✓

Prompt: Can you tell me what my program does?

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 10.55s

Agent Response:

Hey there, Scratch superstar! Let's break down your awesome program: * **`when [green flag] clicked`:** This block is like the "go" button for your code! When you click the green flag, your program starts running. * **`repeat [10] times`:** This block is like a magic loop! It tells the code inside to do the same thing over and over. In your case, it will repeat 10 times. * **`move [10] steps`:** This block tells your sprite to move forward. You've set it to move 10 steps each time. **So, what happens when you click the green flag?** 1. The `repeat` block starts its loop and does everything inside it 10 times. 2. Each time through the loop, your sprite moves 10 steps forward. That means your sprite will move a total of 100 steps (10 steps each time, repeated 10 times). Great job!

Response Verification: Accurate
Route Correct: True

Notes: Contains code explanation, Addresses prompt (2/4 key words match, 50.0%)

Test 4

✗

Prompt: What's happening in my workspace?

Expected Route: code_explain | **Actual Route:** general_agent | **Time:** 7.07s

Agent Response:

Hey there, Scratch explorer! 👋 It sounds like you're curious about your Scratch workspace. That's awesome! To help you, tell me a bit more about what you're wondering: * Do you see any **oops messages**? What do they say? * Is your code doing something different than you thought? Tell me what you wanted to happen and what *is* happening. * Are you stuck on a specific block or a group of blocks? Which one? * Or are you just curious about how the whole Scratch workspace works? The more you tell me, the better I can help you on your Scratch adventure! Let's have fun coding! 🚀

Response Verification: Not Accurate
Route Correct: False
Notes: Wrong route: expected code_explain, got general_agent

Test 5

✓

Prompt: Describe my current Scratch project

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 16.04s

Agent Response:

Hey there, Scratch superstar! Let's break down your awesome project: You've got a cool setup where something happens when you tell it to! 1. **The "When" Block:** This is like the starting gun! It tells your code when to GO! Is it when you click the green flag, press a key, or click on a character? 2. **The "Repeat" Block:** This is like saying, "Do this a bunch of times!" You can tell it how many times to repeat. If you don't say a number, it will repeat 10 times. 3. **The "Move" Block:** This tells your character to move! You can tell it how many steps to take. If you don't say a number, it will move 10 steps. So, your character waits for something to happen, then moves a bit, and does that over and over! **Want to make it even cooler?** * Pick the right "when" block to start your code. * Change the numbers in the "repeat" and "move" blocks to make your character move differently. * Add more blocks to make your character turn, change colors, or make sounds! Keep experimenting and have fun creating! You're doing great!

Response Verification: Accurate
Route Correct: True
Notes: Contains code explanation, Addresses prompt (1/4 key words match, 25.0%)

Test 6 ✓

Prompt: What blocks am I using right now?

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 10.93s

Agent Response:

Hey there, Scratch superstar! Let's look at your awesome code! You've put some cool blocks in the coding area, which starts around the X: 310, Y: 160 spot. Here's what those blocks do: 1. **`when` Block:** This block is like a starting signal. It could be `when green flag clicked`, `when key pressed`, or `when this sprite clicked`. It tells your code when to begin! It's at the top, around X: 342, Y: 124. 2. **`repeat` Block:** This block is like a magic loop! It repeats the code inside it as many times as you want. Find it around X: 342, Y: 157, right below the `when` block. 3. **`move` Block:** This block makes your sprite zoom around! It tells your sprite to move a certain number of steps. It's inside the `repeat` block at X: 342, Y: 243, so your sprite will move again and again! Great job! Your code will make your sprite move repeatedly when the starting signal happens. Keep up the fantastic work!

Response Verification: Accurate
Route Correct: True
Notes: Contains code explanation, Addresses prompt (3/5 key words match, 60.0%)

Test 7 ✓

Prompt: Explain my code to me

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 11.74s

Agent Response:

Hey there, Scratch superstar! Let's see what your code does: 1. **Green Flag Time!** The `when green flag clicked` block is like a starting gun. When you click the green flag, your code comes to life! 2. **Repeat Magic!** The `repeat [10]` block is like a super cool loop. It tells the code inside to do the same thing ten times! 3. **Move, Sprite, Move!** The `move [10] steps` block tells your character to walk forward a little bit. **So, what happens?** When you click the green flag, your sprite will move forward ten steps. But because of the `repeat` block, it does this ten times! That means your sprite moves a lot – like taking ten little walks all in a row! Great job!

Response Verification: Accurate
Route Correct: True
Notes: Contains code explanation, Addresses prompt (1/2 key words match, 50.0%)

Test 8 ✓

Prompt: Can you walk me through what my blocks do?

Expected Route: code_explain | **Actual Route:** code_explain | **Time:** 11.54s

Agent Response:

Hey there, Scratch superstar! Let's see what your cool code does! * **`when [green flag clicked]`:** This block is like the "go" button for your code! When you click the green flag, your Scratch game starts! * **`repeat [10]`:** This block is like a magic loop! It tells the code inside to do the same thing over and over. The number 10 means it will repeat that code ten times! * **`move [10] steps`:** This block tells your sprite to move! The number 10 means it will move ten steps forward each time. **How it all works:** 1. You click the green flag, and the game starts! 2. The `repeat` block says, "Okay, let's do this ten times!" 3. Each time, the sprite will `move 10 steps`. 4. After moving, the loop starts again until it has moved ten times! **So, when you click the green flag, your sprite will move a lot! It will move 10 steps, ten times, which is 100 steps in total!**

Response Verification: Accurate

Route Correct: True
Notes: Contains code explanation, Addresses prompt (1/4 key words match, 25.0%)

Code Debugging Tests

Test 1 ✓

Prompt: My code isn't working, can you help?

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 12.41s

Agent Response:

Hey there, Scratch superstar! Let's make sure your blocks are working together to create something amazing! * **"When" Block:** I see you have a "when" block, which is awesome because that's how your code knows when to start! But, there are many "when" blocks. Let's use "when green flag clicked", "when key pressed", "when this sprite clicked", "when backdrop switches", "when greater than", or "when I receive message". * **"Repeat" Block:** You also have a "repeat" block, which is like saying, "Do this a bunch of times!" To work, it needs to be right after the "when" block. * **"Move" Block:** And a "move" block! This tells your sprite to move. To make it move again and again, let's put it inside the "repeat" block. **So, here's the plan:** 1. Drag the "repeat" block under the "when" block. It's like connecting train cars! 2. Now, drag the "move" block inside the "repeat" block. This tells the sprite to move again and again! **Why do we do this?** The blocks need to be in the right order, like a recipe! The "when" block starts the action, the "repeat" block says how many times to do it, and the "move" block says what to do! If they're not connected, your code won't know what to do. Keep up the great work!

Response Verification: Accurate

Route Correct: True
Notes: Contains debugging advice, Addresses prompt (1/4 key words match, 25.0%)

Test 2 ✓

Prompt: Why isn't my sprite moving?

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 11.96s

Agent Response:

Hey there, Scratch coder! Let's get your sprite moving! Here's what I see in your code: * The `when green flag clicked` block is like the "go" button for your code! * The `repeat 10` block is like saying, "Do this 10 times!" * The `move 10 steps` block is what tells your sprite to walk. But, uh-oh! It looks like your sprite isn't listening to you because the `move 10 steps` block isn't *inside* the `repeat 10` block. It's like telling someone to walk, but not telling them *how many times* to walk! Here's how to fix it: 1. Drag the `move 10 steps` block *inside* the `repeat 10` block. You'll see the `repeat` block light up when it's ready to "catch" the `move` block. Now, when you click the green flag, your sprite will move 10 steps, and it will do that 10 times! Yay! If it *still* doesn't work, don't worry! We'll figure it out together! Just let me know!

Response Verification: Not Accurate

Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 3 ✓

Prompt: There's something wrong with my program

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 10.89s

Agent Response:

Hey there, Scratch superstar! I've checked out your awesome project, and it looks like you're about to make something super cool happen! Here's the lowdown: * You've got a "when green flag clicked" block. That's the perfect way to kick off your program when you press the green flag! * You also have a "repeat 10" block. This is like saying, "Do this stuff 10 times!" * And a "move 10 steps" block to make your character move. **Uh-oh!** It looks like your "move 10 steps" block isn't quite inside the "repeat 10" block yet. That means your character won't move 10 steps, 10 times. **Let's fix it!** Drag the "move 10 steps" block into the "repeat 10" block. It's like giving the "repeat" block a hug! **Ta-da! ** Now, when you click the green flag, your character will move 10 steps, and it will do it 10 times in a row! How cool is that? You're doing great!

Response Verification: Not Accurate
Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 4 ✓

Prompt: Help me fix this bug

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 10.43s

Agent Response:

Hey there, Scratch superstar! I looked at your code, and it's almost perfect! * The `when green flag clicked` block is great. It tells your code to start when you click the green flag. * The `repeat 10` block is super cool. It's like saying, "Do this 10 times!"

* The `move 10 steps` block makes your character move forward. Here's a tiny thing to fix: Right now, the `move 10 steps` block isn't *inside* the `repeat 10` block. That means your character will only move once. **Here's how to make it awesome:** Drag the `move 10 steps` block *inside* the `repeat 10` block. It's like putting a toy inside a box. Now, your character will move 10 steps, and it will do that 10 times! Yay!

Response Verification: Not Accurate
Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 5

✓

Prompt: My blocks aren't doing what I expected

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 15.7s

Agent Response:

Hey there, Scratch superstar! I've looked at your awesome project, and I have a super simple tip to make it even better! You have a "when green flag clicked" block - that's perfect for starting your code! You also have a "repeat (10)" block, which is great for doing something again and again. And I see a "move (10) steps" block to make your character move. Here's the trick: The "move (10) steps" block needs to go INSIDE the "repeat (10)" block. Think of it like putting toys inside a box! * Drag the "move (10) steps" block up into the "repeat (10)" block. * You'll see a white line show you where it will go. Let the "move" block go, and it will snap right inside! Now, when you click the green flag, your character will move ten times! Awesome!

Response Verification: Not Accurate
Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 6 ✓

Prompt: Debug my Scratch project

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 10.61s

Agent Response:

Hey there, Scratch superstar! I looked at your code, and it's almost perfect! Here's a tiny tweak to make it even better: You have a "when green flag clicked" block—that's great for starting your project! You also have a "repeat 10" block, which is awesome for making things happen again and again. And I see you have a "move 10 steps" block to make your character move. But, the "move 10 steps" block needs to be *inside* the "repeat 10" block. Think of it like this: the "repeat" block is like a loop, and you want the "move" block to be inside the loop so it happens over and over. Just drag the "move 10 steps" block into the "repeat 10" block. Now, when you click the green flag, your character will move ten steps, ten times! Keep up the fantastic work!

Response Verification: Not Accurate
Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 7 ✓

Prompt: What's wrong with my code?

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 10.33s

Agent Response:

Hey there, Scratch superstar! I've checked out your awesome project, and I have a super easy tip to make it even better! I see you have these blocks: * The **`when green flag clicked`** block: This is like the "start" button for your code! * The **`repeat (10)`** block: This block is like saying, "Do this 10 times!" * The **`move

(10) steps`** block: This tells your character to move a little bit. Here's the trick: The `move` block needs to go INSIDE the `repeat` block. It's like putting a toy inside a box! **Here's how:** 1. Click and drag the `move (10) steps` block. 2. Move it over the `repeat (10)` block. You'll see a white line show you where it will go. 3. Let go of the mouse, and *poof*, it's inside! Now, when you click the green flag, your character will move 10 steps, and it will do that 10 times in a row! Awesome, right? Keep up the fantastic work!

Response Verification: Not Accurate
Route Correct: True
Notes: Missing debugging analysis or solutions, Response doesn't sufficiently address prompt or route requirements

Test 8

✓

Prompt: My sprite keeps disappearing

Expected Route: code_debugging | **Actual Route:** code_debugging | **Time:** 12.97s

Agent Response:

Hey there! I'm here to help you find your missing sprite! It's like a fun detective game! Here's what we can do: 1. **Make Sure Your Sprite Can Be Seen:** * Sometimes, sprites hide and forget to come back! Let's add a "show" block right after the green flag. It's like telling your sprite, "Time to shine!" * Check if there is a "hide" block that might be making your sprite disappear. 2. **Keep Your Sprite on the Stage:** * Your sprite might be running off the stage where we can't see it. * Let's add a "if on edge, bounce" block inside the "repeat" loop. It's like a little bumper that says, "Oops, turn back!" * Or, we can tell your sprite to take smaller steps so it doesn't run away too fast. Let's put these blocks together like this: ```scratch when [green flag clicked] show repeat [10] move [10] steps if on edge, bounce ``` Try these tricks, and let me know if your sprite is back! If not, we'll keep searching together!

Response Verification: Accurate
Route Correct: True

Notes: Contains debugging advice, Addresses prompt (1/3 key words match, 33.3%)

Give Instructions Tests

Test 1

✓

Prompt: How do I make a sprite move?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 14.66s

Agent Response:

Hey there, future Scratch superstar! Let's make your characters zoom around the screen! **1. Simple Steps:** * Find the blue "move steps" block. It tells your character to walk! * Put it inside an orange "forever" block to keep moving. * Add a yellow "when green flag clicked" block on top to start when you press the green flag. Like this: ```scratch when green flag clicked forever move 10 steps ``` Want to bounce when you hit the edge? Add the "if on edge, bounce" block! ```scratch when green flag clicked forever move 10 steps if on edge bounce ``` **2. Arrow Key Control:** * Use the yellow "when key pressed" block for each arrow key. * Use the blue "change x by" block to move left and right. Positive numbers go right, negative go left! * Use the blue "change y by" block to move up and down. Positive numbers go up, negative go down! * Use the blue "point in direction" block to make the sprite face the correct direction. ```scratch when key pressed [right arrow v] change x by 10 when key pressed [left arrow v] change x by -10 when key pressed [up arrow v] change y by 10 when key pressed [down arrow v] change y by -10 ``` **3. Random Fun:** * Use the blue "go to random position" block. * Put it in a "forever" loop to keep moving randomly. * Add a "wait 1 seconds" block to slow it down. ```scratch when green flag clicked forever go to random position wait 1 seconds ``` Have fun making your sprites move! You're doing great!

Response Verification: Accurate

Route Correct: True

Notes: Contains instructional content, Addresses prompt (2/3 key words match, 66.7%)

Test 2 ✓

Prompt: Teach me how to use repeat blocks

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 12.68s

Agent Response:

Hey there, Scratch explorer! Let's learn about the "repeat" block. It's like a magic trick to make things happen again and again! **What it does:** The "repeat" block does the same thing many times. **How to use it:** 1. **Find it:** Go to the **Control** blocks. You'll see the "repeat" block. 2. **Drag it:** Bring the "repeat" block to your workspace. 3. **Set the number:** Type a number in the white box. This tells Scratch how many times to repeat. Like "5" or "10"! 4. **Add blocks inside:** Put the blocks you want to repeat *inside* the "repeat" block. 5. **Start it:** Use a "when green flag clicked" block from **Events** to start the repeating. **Example:** Let's make a sprite move 10 steps forward 5 times. 1. Get a "when green flag clicked" block. 2. Add a "repeat" block and type "5" inside. 3. Put a "move 10 steps" block (from **Motion**) *inside* the "repeat" block. Now, when you click the green flag, your sprite will move 10 steps, then do it again, and again, until it has moved 5 times! **Another Example: Changing Color** Let's change the color of a sprite 20 times. 1. Get a "when green flag clicked" block. 2. Add a "repeat" block and type "20" inside. 3. Put a "change color effect by 25" block (from **Looks**) *inside* the "repeat" block. The color will change 20 times. Have fun repeating!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (2/3 key words match, 66.7%)

Test 3 ✓

Prompt: What's the best way to make animations?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 13.93s

Agent Response:

Hey there, future animators! Let's learn how to make things move in Scratch! **1. Costume Magic (Like a Flipbook!):** * Imagine you're making a flipbook. You draw the same thing over and over, but change it a little bit each time. In Scratch, we use **costumes** to do the same thing! * **What to use:** * `next costume`: This block changes your character to the next picture. * `wait [number] seconds`: This block makes Scratch wait a little bit before changing to the next picture. * `forever`: This block makes your animation keep going and going! * **Example:** ```scratch when green flag clicked forever next costume wait 0.1 seconds end ``` **2. Moving Around:** * Instead of changing costumes, we can make our character move across the screen! * **What to use:** * `move [number] steps`: This block makes your character take steps. * `change x by [number]`: This block moves your character left or right. * `change y by [number]`: This block moves your character up or down. * **Example:** ```scratch when green flag clicked forever change x by 10 wait 0.05 seconds if <(x position) > (240)> then set x to (-240) end end ``` **3. Super Animation Combo!** * Want to be a super animator? Use both costumes and moving around! Imagine a character walking - you'd change their leg costumes *and* move them forward! **Tips for Awesome Animations:** * Start simple! * Plan what you want to make. * Have fun experimenting! * Use more than one character! You got this! Now go make some amazing animations!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (1/4 key words match, 25.0%)

Test 4 ✓

Prompt: How can I make my sprite jump?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 13.13s

Agent Response:

Hey there, future game developers! Let's make your Scratch sprite jump! It's super fun! **First, pick how you want to make your sprite jump:** * Press a key? Use the `when key pressed` block. You can find it in the "Events" blocks. Choose your favorite key, like the space bar! * Click the sprite? Use the `when this sprite clicked` block, also in "Events." **Now, let's make the sprite jump up and down:** 1. **Jump Up:** Use the `change y by` block from the "Motion" blocks. Put in a positive number, like 50, to make it go up! 2. **Wait at the Top:** Add a `wait seconds` block from the "Control" blocks. A short wait, like 0.2 seconds, will do! 3. **Come Down:** Use another `change y by` block, but this time with a negative number, like -50. This brings the sprite back down! **Make it even cooler with a loop!** 1. Use the `repeat` block from the "Control" blocks. 2. Put `change y by (10)` and `wait (0.01) seconds` inside the `repeat` block. 3. Repeat with `change y by (-10)` and `wait (0.01) seconds` to come down smoothly. Try different numbers to make the jump higher or lower. Have fun experimenting!

Response Verification: Accurate

Route Correct: True

Notes: Contains instructional content, Addresses prompt (2/3 key words match, 66.7%)

Test 5 ✓

Prompt: Explain how to use if-then blocks

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 11.08s

Agent Response:

Hey there, Scratch Explorers! Want to make your Scratch games super smart? Let's learn about the `if then` block! It's like a brain for your code, helping it make choices. **What it does:** The `if then` block checks if something is true. If it is, then the code inside the block runs! If not, Scratch skips that code. **How to use it:** 1. Find the `if then` block in the "Control" blocks (it's at the bottom of the list). Drag it into your code. 2. Now, we need to tell the `if then` block *what* to check. We need to put a condition in the hexagonal space. You can find conditions in "Sensing" or "Operators." * `touching object`: Is my sprite touching something? * `key pressed`: Is someone pressing a key? * `greater than`, `less than`, `equals`: Are two numbers the same? 3. Put the code you want to run *inside* the `if then` block. This code will only run if your condition is true! **Example:** Let's make a sprite say "Hello!" when you press the space key. 1. Get a `when key pressed` block from the "Events" blocks. Choose "space." 2. Put an `if then` block inside the `when key pressed` block. 3. Drag a `key pressed` block from "Sensing" into the `if then` block's hexagon. Choose "space" again. 4. Put a `say message` block inside the `if then` block. Change the message to "Hello!". Now, your sprite will only say "Hello!" when you press the space key! Isn't that cool? The `if then` block helps you make games that react to what's happening. Have fun experimenting!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (1/3 key words match, 33.3%)

Test 6 ✓

Prompt: How do I add sound to my project?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 10.67s

Agent Response:

Hey there, future sound designers! Let's add some awesome sounds to your Scratch project! 1. **Find the Sound Blocks:** Look on the left side of your screen for the "Sound" button. Click it, and you'll see all the sound blocks you can use! 2. **Pick a Sound Block:** * `play sound until done`: This block plays the whole sound before moving on. * `start sound`: This block starts the sound, but keeps going with the code right away. Great for background music! * `stop all sounds`: This block stops all the sounds. 3. **Choose When to Play the Sound:** * `when green flag clicked`: The sound plays when you start the project. * `when key pressed`: The sound plays when you press a key. * `when this sprite clicked`: The sound plays when you click on a sprite. 4. **Put the Blocks Together:** Drag the "when" block to the middle. Then, drag your sound block and snap it under the "when" block. 5. **Pick Your Sound:** Click the little arrow on the sound block to pick a sound. Scratch has lots of sounds to choose from! You can even add your own sound by clicking "Choose a Sound" in the "Sounds" tab! **Example:** To play a sound when the green flag is clicked: * Use `when green flag clicked` * Use `play sound until done` Snap them together and pick your sound. Now, click the green flag and hear your sound play! You did it!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (1/2 key words match, 50.0%)

Test 7 ✓

Prompt: What are the steps to create a game?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 16.42s

Agent Response:

Hey there, future game developer! Let's make a Scratch game together! 1. **Dream It, Plan It:** * What kind of game do you want? A jumping game? A puzzle? * What will players do? What are the rules? * Draw your game! What will it look like? Who are the characters? 2. **Set the Stage:** * **Pick a Backdrop:** Go to "Looks" and pick a cool background! Use the `switch backdrop to` block to show it. * **Add Sprites:** These are your characters and things in the game. You can draw them or pick from Scratch's library! 3. **Make It Move:** * **Start Button:** Use `when green flag clicked` (in "Events") to start the game. * **Move Around:** Use "Motion" blocks to move your sprites! * `move [10] steps` makes it move forward. * `turn right [15] degrees` makes it turn. * **Use the Keyboard:** Use `when key pressed` (in "Events") to move with the arrow keys! 4. **Make It Interact:** * **Touching:** Use `touching [object v]` (in "Sensing") to see when sprites bump into each other. * **If...Then:** Use `if then` (in "Control") to make things happen when they touch! 5. **Keep Score:** * **Make a Variable:** In "Variables," make a box to hold the score. * **Add Points:** Use `change [score v] by [1]` to add to the score. * **Show the Score:** Use `show variable [score v]` to show it on the screen. 6. **Add Sounds:** * **Pick Sounds:** Go to "Sound" and choose fun noises! * **Play Sounds:** Use `start sound [sound name v]` to play them when things happen. 7. **Repeat Actions:** * **Forever:** Use `forever` (in "Control") to keep checking things. * **Repeat:** Use `repeat [10]` to do something a few times. 8. **Game Over:** * **When to Stop:** Use `if then` to check if the game is over (like if you run out of lives). * **Stop It:** Use `stop [all v]` (in "Control") to end the game. 9. **Make It Awesome:** * **Costumes:** Change how your sprites look with "Looks"! * **Messages:** Use `broadcast [message v]` (in "Events") to let sprites talk to each other. Start small and add more cool stuff as you go! Have fun!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (2/4 key words match, 50.0%)

Test 8

✓

Prompt: How do variables work in Scratch?

Expected Route: give_instructions | **Actual Route:** give_instructions | **Time:** 12.03s

Agent Response:

Hey there, Scratch Explorers! Variables are like special boxes that hold secrets in your Scratch games. Let's learn how to use them! 1. **Make a Box (Variable):** * Go to "Variables" and click "Make a Variable." * Give it a cool name like "score" or "speed." 2. **Put Something in the Box:** * Use "set variable to" to give your box a starting value. * Like, `set [score v] to [0]` to start the score at zero! 3. **Change What's in the Box:** * Use "change variable by" to add or subtract from the value. * `change [score v] by [1]` to add one to the score! 4. **Use the Box's Secret:** * Drag your variable into other blocks. * `move [speed v] steps` makes your character move faster or slower. * `say [join[Score: ][score v]]` to show the score on the screen! 5. **Show or Hide the Box:** * Use "show variable" and "hide variable" to make the value visible or invisible on the stage. **Example:** 1. Make a "score" variable. 2. Set "score" to 0 at the start. 3. Add 1 to "score" when the player gets a point. 4. Show the "score" on the stage! Variables help you make awesome, interactive games. Have fun experimenting!

Response Verification: Accurate
Route Correct: True
Notes: Contains instructional content, Addresses prompt (1/3 key words match, 33.3%)

General Agent Tests

Test 1 ✓

Prompt: Hello!

Expected Route: general_agent | **Actual Route:** general_agent | **Time:** 5.97s

Agent Response:

Hello! I'm here to help you learn Scratch in a fun and easy way. If you have a Scratch project you want to make, or if you're stuck on a step, just tell me. I can help you break it down into simple steps that are easy for kids to follow. Let's have fun creating with Scratch! What do you want to make today?

Response Verification: Accurate
Route Correct: True
Notes: General response provided, Addresses prompt (1/1 key words match, 100.0%)

Test 2 ✓

Prompt: Thank you for your help

Expected Route: general_agent | **Actual Route:** general_agent | **Time:** 5.45s

Agent Response:

Yay! I'm super happy I could help you with your Scratch project! If you need anything else, just ask. Have lots of fun Scratching and creating awesome things! You're doing great!

Response Verification: Accurate
Route Correct: True
Notes: General response provided, Addresses prompt (2/3 key words match, 66.7%)

Test 3 ✓

Prompt: What is Scratch?

Expected Route: general_agent | **Actual Route:** general_agent | **Time:** 6.49s

Agent Response:

Hey there, future coder! Scratch is like a super fun playground where you can make your own games, stories, and cartoons! Instead of typing complicated code, you get to use colorful blocks that snap together like LEGOs. Each block tells your characters what to do. It's like giving them instructions with puzzle pieces! You can drag and drop these blocks to make your characters move, talk, and even dance. It's super easy to learn, and you'll be amazed at what you can create. Once you've made something awesome, you can share it with other kids in the Scratch community. You can also check out their projects and get inspired. It's a great way to learn and make new friends! So, get ready to unleash your imagination and start creating with Scratch!

Response Verification: Accurate
Route Correct: True
Notes: General response provided, Addresses prompt (1/2 key words match, 50.0%)

Test Environment

Testing Script: AgentTesting.py

Results Directory: agent_test_results/

Chrome Debug Port: 9222

Stricter Accuracy Criteria: Wrong route = Not Accurate, Response must match route requirements and prompt context (30%+ keyword overlap)